

KASSEL, Germany

WMO: 104380

Lat: 51.296N

Lon: 9.443E

Elev: 518

StdP: 14.42

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	10.1	15.1	4.3	7.0	12.9	8.5	8.7	17.3	22.9	42.8	20.5	38.9	5.7	0	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	83.2	65.8	79.9	64.1	76.9	62.8	67.8	78.8	65.9	75.9	64.3	73.7	7.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	90.3	71.7	62.2	85.2	70.2	60.6	80.6	68.2	32.4	78.9	30.9	76.1	29.7	73.6	74.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.7	17.2	15.2	DB	8.8	89.9	8.1	3.1	3.0	92.2	-1.8	94.0	-6.3	95.7	-12.2	98.0	(4)
(5)			WB	8.2	71.6	8.2	1.8	2.3	72.8	-2.5	73.9	-7.1	74.9	-13.0	76.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.1	32.6	32.4	40.2	46.0	54.8	59.7	64.0	63.8	57.0	49.4	40.0	35.7		
	DBStd	13.20	9.27	8.55	6.71	6.58	6.75	6.46	5.88	5.74	5.42	6.26	6.52	6.85		
	HDD50	2369	540	495	310	155	29	3	0	0	5	86	303	443		
	HDD65	6338	1005	914	770	571	321	183	93	95	245	483	751	907		
	CDD50	1663	0	1	5	34	178	295	435	427	216	69	2	1		
	CDD65	158	0	0	0	0	5	26	63	57	6	1	0	0		
	CDH74	1345	0	0	0	0	59	219	539	476	51	1	0	0		
	CDH80	277	0	0	0	0	2	36	119	118	2	0	0	0		
(14) Wind	WSAvg	6.6	8.8	7.4	7.3	6.5	5.7	5.5	5.2	5.1	5.7	6.3	7.4	8.0		
(15) Precipitation	PrecAvg	27.8	2.1	1.7	2.0	2.0	2.7	3.0	2.7	2.6	2.2	2.0	2.4	2.6		
	PrecMax	42.8	5.5	4.9	4.3	5.2	6.0	6.7	7.3	6.0	5.4	6.9	5.0	4.9		
	PrecMin	19.6	0.1	0.2	0.4	0.4	0.4	0.7	0.6	0.7	0.5	0.4	0.2	0.2		
	PrecStd	5.4	1.3	1.1	1.0	0.9	1.3	1.3	1.5	1.2	1.2	1.4	1.0	1.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.8	55.6	63.7	69.9	79.3	84.2	87.3	88.8	79.4	70.4	55.7	53.3		
		MCWB	47.4	49.5	50.8	54.1	60.9	67.5	67.1	67.9	62.1	59.8	51.0	49.7		
	2%	DB	47.7	50.2	57.0	65.6	75.3	79.8	83.4	83.1	74.7	64.8	52.7	50.4		
		MCWB	45.0	46.1	47.9	52.3	58.8	64.2	65.9	65.4	62.1	57.7	49.6	47.5		
	5%	DB	45.7	46.5	52.7	61.6	71.6	75.9	80.0	79.1	70.5	61.1	50.8	48.0		
		MCWB	43.3	42.9	46.4	50.3	57.2	62.1	64.8	63.9	60.4	55.6	48.1	45.8		
	10%	DB	43.6	42.8	49.4	57.5	68.1	72.0	76.6	75.5	66.9	58.2	48.7	45.2		
		MCWB	41.3	39.7	44.6	48.1	56.1	60.5	63.4	62.8	59.0	53.6	46.1	43.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	50.3	52.0	56.0	63.0	69.7	70.0	70.2	65.8	61.3	52.4	50.2		
		MCDB	49.9	54.6	60.1	67.0	73.8	81.5	82.4	83.1	72.9	67.9	53.7	52.0		
	2%	WB	45.7	47.0	49.5	53.7	60.7	66.2	67.9	67.1	63.5	58.5	50.4	47.8		
		MCDB	47.4	50.2	55.8	63.2	71.5	75.5	79.1	78.5	70.7	63.4	52.3	49.8		
	5%	WB	43.6	43.2	47.4	51.4	58.7	63.8	66.1	65.4	61.6	56.2	48.6	45.7		
		MCDB	45.3	45.8	51.9	59.2	69.3	73.0	76.4	75.6	68.4	60.1	50.5	47.7		
	10%	WB	41.4	40.2	45.2	49.3	56.7	61.8	64.3	63.9	59.7	54.3	46.4	43.7		
		MCDB	43.5	42.6	48.9	56.0	66.9	70.1	74.0	73.1	65.8	57.5	48.3	45.3		
(35) Mean Daily Temperature Range	MDBR		7.3	9.7	12.1	15.1	16.6	15.6	16.8	17.5	14.7	12.1	8.3	6.9		
	5% DB	MCDBR	8.3	13.1	18.4	21.4	22.5	22.3	23.0	24.1	21.4	16.2	10.3	8.9		
		MCWBR	7.6	9.6	11.2	10.7	10.6	9.9	9.7	10.0	10.3	9.5	8.1	7.8		
	5% WB	MCDBR	8.3	12.8	16.5	18.6	20.7	19.8	19.6	20.7	18.5	14.5	9.8	8.9		
		MCWBR	8.0	9.9	11.0	10.6	10.6	9.9	9.6	9.6	9.8	8.9	8.3	8.0		
(40) Clear-Sky Solar Irradiance	taub		0.313	0.338	0.374	0.414	0.418	0.416	0.429	0.423	0.387	0.358	0.337	0.310		
	taud		2.390	2.349	2.271	2.200	2.224	2.251	2.219	2.258	2.342	2.410	2.399	2.385		
	Ebn at Noon		223	248	259	261	266	267	260	255	252	237	211	204		
	Edh at Noon		20	26	34	41	42	42	42	38	32	25	19	17		
(44) All-Sky Solar Radiation	RadAvg		223	439	790	1263	1514	1655	1570	1343	966	565	253	167		
	RadStd		33	69	108	157	163	152	208	125	134	59	38	20		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (42 neighbors)		+0.58	N/A	N/A	N/A	N/A	N/A	N/A	-216	N/A	N/A	

Nomenclature: See separate page